



ITEC 204 – Data Structures and Algorithms

Name: Nombrado, Georgia Mae G.	Date: September 25, 2026
Section: BSIT_2B	Instructor: Maxyne Nuela Ignacio

- I. **Objective:** Write a short explanation of what your program is designed to do. (Minimum of 3 sentences)

Answer: The program is designed to create an IT Automation Incident Ticket Manager that organizes and manages incident tickets generated by automation bots. It allows users to add, display, search and remove incident tickets using their incident ID, bot name, and short description. The program also counts the total number of active incident tickets, helping support teams manage and monitor incidents efficiently.

- II. **Program Operations:** Briefly explain how your program performs each operation. (Insert Specific Code Line and Your Explanation)

Add Incident:

```
new_ticket = {  
    "Id": incident_id,  
    "bot": bot,  
    "description": description  
}  
tickets.append(new_ticket)
```

This code adds a new incident ticket to the end of the tickets list using append(). Each ticket includes incident ID, bot name and short description.

Display Incidents:

```
for ticket in tickets:  
    print(f'| {ticket["id"]:<13} | {ticket["bot"]:<18} | {ticket["description"]:<59} |')
```

This loop goes through the tickets list and displays each ticket's ID, bot name and short description.

Search Incident:

```
for ticket in tickets:  
    if ticket["id"] == incident_id:  
        print("Ticket Found")  
        return
```

This code loops through the list to find a ticket matching the user's incident ID and displays it once found.



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Remove Resolved Incident:

for ticket in tickets:

 if ticket["id"] == incident_id:

 tickets.remove(ticket)

 return

This code finds the user's incident ID and uses remove() to delete the resolved ticket from the list.

Count Active Incidents:

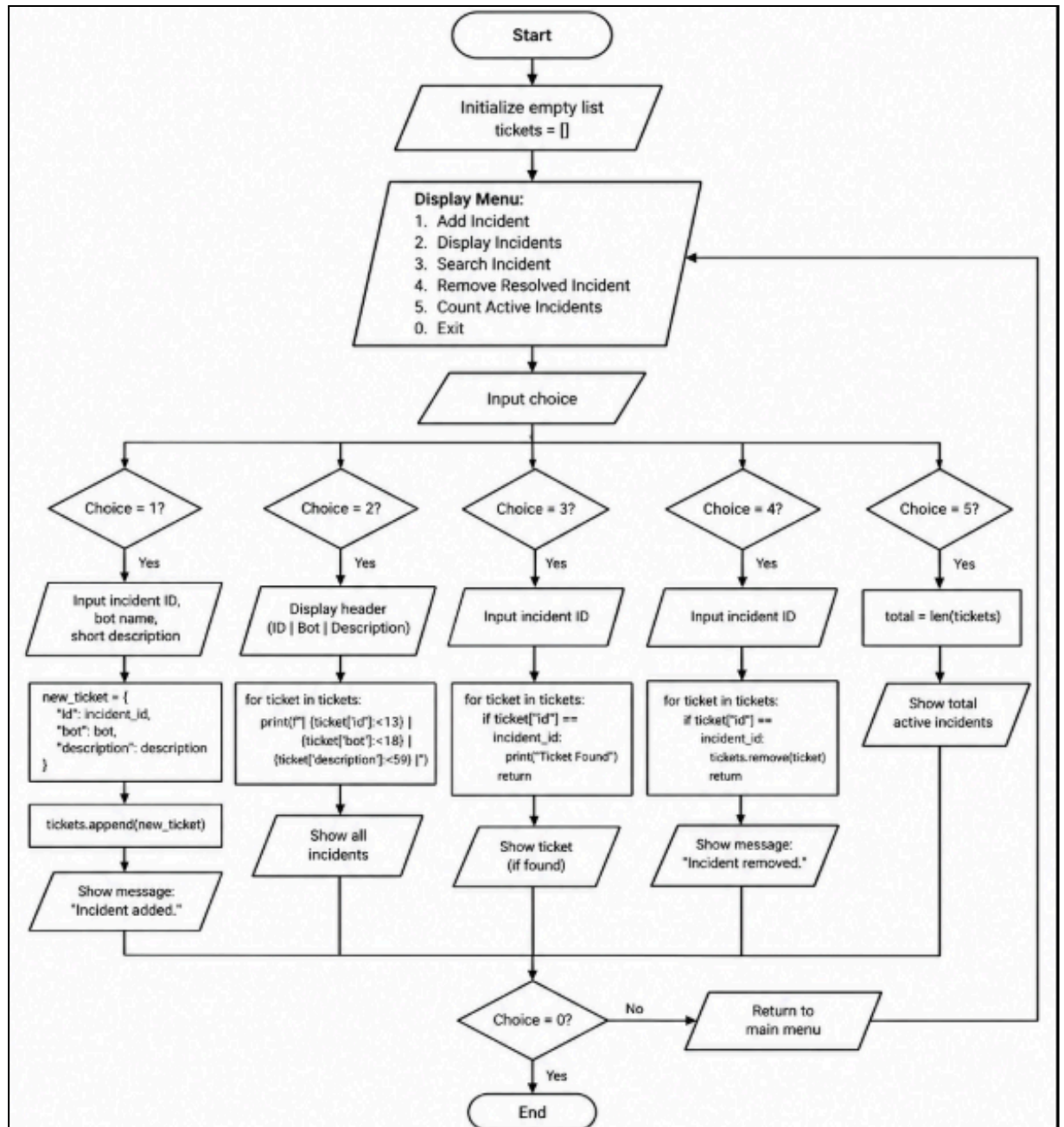
total = len(tickets)

The len() function counts the tickets in the list to get the total number of active incidents.



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III. **Flowchart:** Create a flowchart showing the overall process and logic of your program.



IV. **Conclusion:** Briefly explain what you learned from the activity and how the linear data structure was applied.

In this activity, I learned how to use a linear data structure, specifically a list, to organize and manage incident tickets. I understood how different operations, such as `append()`, `remove()`, looping, and `len()`, can be used to add, display, search, remove, and count tickets. This activity helped me understand how lists can be applied in real-world IT situations, such as managing incident records and keeping track of active tickets in an organized and efficient way. Additionally, I also know the importance of making a flowchart.